LEO PROTOTYPE OVERVIEW
Institutional Integrity Infrastructure Prototype

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Project: LEO – Institutional Integrity Infrastructure

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1. Project Status

LEO is a **functional prototype system** designed to analyze institutional processes and detect structural anomalies in financial, governance and administrative systems.

The platform is currently being developed and tested within the **BBS Foundation institutional sandbox environment**.

The project is **not a conceptual proposal**.

Core system components have already been implemented and tested.

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2. Implemented Prototype Components

The current prototype includes the following operational subsystems:

Integrity Runtime Engine

- institutional event processing
- anomaly detection pipeline
- investigation and escalation logic

Epistemic Truth Layer

- structured truth ingestion engine
- append-only epistemic memory
- verifiable state roots

Distributed Verification Network

- multi-core consensus model
- fork detection and integrity verification
- network simulation environment

Investigation and Escalation Pipeline

- anomaly clustering
- investigation report generation
- case escalation and institutional alert system

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3. Example Prototype Output

Example output produced by the system during testing:

CASE ESCALATION RESULT


```
{
  "action": "CRITICAL_ESCALATION",
  "alert": {
    "status": "INSTITUTIONAL_ALERT_CREATED"
  },
  "case": {
    "status": "CASE_CREATED"
  }
}
```


This demonstrates the ability of the system to detect anomalies and generate institutional escalation events.

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4. Current Development Focus

Ongoing development focuses on:

- financial integrity analysis modules
- governance and election integrity verification
- distributed verification infrastructure
- institutional anomaly investigation tools

The system continues to evolve independently as an experimental governance infrastructure.

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5. Research and Collaboration

The BBS Foundation welcomes collaboration with:

- academic institutions
- research laboratories
- public governance organizations

The LEO platform is intended as a **research infrastructure for institutional integrity systems**.

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6. Repository and Architecture Overview

System architecture and repository structure:

https://leonexus.org

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2. Live Runtime Trace (PowerShell Output)

Note: The following is a raw capture from the development environment.

```
Windows PowerShell

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Try the new cross-platform PowerShell https://aka.ms/pscore6


PS C:\Users\user>  cd D:\BBS-09-01-2026

PS D:\BBS-09-01-2026> python test_anomaly_memory.py

{'status': 'ANOMALY_RECORDED', 'file': 'epistemic_v1_storage/anomaly_library.jsonl'}

{'total_anomalies': 1}

Recent anomalies: [{'record_type': 'ANOMALY', 'timestamp': '2026-03-08T08:20:06.346267',
'anomaly': {'type': 'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'deviation': 2.7,
'contract_id': 'PROC-77821'}}]

PS D:\BBS-09-01-2026> python test_anomaly_pattern_engine.py

{'patterns_detected': 0, 'patterns': []}

PS D:\BBS-09-01-2026> python test_anomaly_memory.py

C:\Users\user\AppData\Local\Programs\Python\Python313\python.exe: can't open file 'D:\\BBS-09-01-
2026\\test_anomaly_memory.py': [Errno 2] No such file or directory

PS D:\BBS-09-01-2026> python test_anomaly_memory.py

{'status': 'ANOMALY_RECORDED', 'file': 'epistemic_v1_storage/anomaly_library.jsonl'}

{'total_anomalies': 2}

Recent anomalies: [{'record_type': 'ANOMALY', 'timestamp': '2026-03-08T08:20:06.346267',
'anomaly': {'type': 'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'deviation': 2.7,
'contract_id': 'PROC-77821'}}, {'record_type': 'ANOMALY', 'timestamp': '2026-03-
08T08:46:38.835975', 'anomaly': {'type': 'PRICE_ANOMALY', 'domain': 'finance', 'severity':
'HIGH', 'deviation': 2.7, 'contract_id': 'PROC-77821'}}]

PS D:\BBS-09-01-2026> python test_anomaly_memory.py

{'status': 'ANOMALY_RECORDED', 'file': 'epistemic_v1_storage/anomaly_library.jsonl'}

{'total_anomalies': 3}

Recent anomalies: [{'record_type': 'ANOMALY', 'timestamp': '2026-03-08T08:20:06.346267',
'anomaly': {'type': 'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'deviation': 2.7,
'contract_id': 'PROC-77821'}}, {'record_type': 'ANOMALY', 'timestamp': '2026-03-
08T08:46:38.835975', 'anomaly': {'type': 'PRICE_ANOMALY', 'domain': 'finance', 'severity':
'HIGH', 'deviation': 2.7, 'contract_id': 'PROC-77821'}}, {'record_type': 'ANOMALY', 'timestamp':
'2026-03-08T08:47:13.254040', 'anomaly': {'type': 'PRICE_ANOMALY', 'domain': 'finance',
'severity': 'HIGH', 'deviation': 2.7, 'contract_id': 'PROC-77821'}}]

PS D:\BBS-09-01-2026> python test_anomaly_pattern_engine.py

{'patterns_detected': 1, 'patterns': [{'pattern_type': 'PRICE_ANOMALY', 'domain': 'finance',
'severity': 'HIGH', 'occurrences': 3}]}

PS D:\BBS-09-01-2026> python -m tests_runtime.test_epistemic_graph

{'edge_hash': 'fd8aa03cb9947d9cd480958a65dad2c6311b00d371d463ee50b9872a4eb38e4c', 'edge':
{'subject': 'Company:f89765700d4f42d2932c2cdfe363e4bb', 'relation': 'OWNS', 'object':
'Company:7256eabaae37459e9c780b0af7fd2c61', 'timestamp': '2026-03-08T09:26:43.859328',
'source_fact_hash': None}}
```

```
TOTAL EDGES: 1
PS D:\BBS-09-01-2026> python -m tests_runtime.system_runtime_audit

LEO RUNTIME STRUCTURE AUDIT

PYTHON MODULES FOUND:

leo\runtime\engine.py
leo\runtime\audit.py
leo\runtime\epistemology.py
leo\runtime\canon_rules.py
leo\runtime\leo_types.py
leo\runtime\audit_verifier.py
leo\runtime\tamper_alerts.py
leo\runtime\event_replay.py
leo\runtime\replay_gate.py
leo\runtime\runtime_flow.py
leo\runtime\runtime_cli.py
leo\runtime\system_boot_check.py
leo\runtime\__init__.py
leo\runtime\runtime_context.py
leo\runtime\state\state_commit.py
leo\runtime\epistemology\epistemic_gatekeeper.py
leo\runtime\epistemology\truth_state.py
leo\runtime\epistemology\epistemic_memory.py
leo\runtime\epistemology\epistemic_certificate.py
leo\runtime\epistemology\epistemic_persistence.py
leo\runtime\epistemology\__init__.py
leo\runtime\epistemology\truth_ingestion_engine_v1.py
leo\runtime\accounting\accounting_entry.py
leo\runtime\accounting\__init__.py
leo\runtime\governance\__init__.py
leo\runtime\governance\governance_entry.py
leo\runtime\governance\integrity_debt.py
leo\runtime\persistence\state_schema.py
leo\runtime\persistence\state_writer.py
leo\runtime\persistence\persistence_coordinator.py
leo\runtime\epoch\execution_epoch.py
leo\runtime\crypto\__init__.py
```

leo\runtime\crypto\signing.py

leo\runtime\crypto\state_hasher.py

leo\runtime\ingestion__init__.py

leo\runtime\ingestion\truth_sources.py

leo\runtime\ingestion\ingestion_cli.py

leo\runtime\ingestion\ingestion_service.py

leo\runtime\ingestion_legacy_v01_ingestion\truth_schema.py

leo\runtime\ingestion_legacy_v01_ingestion\truth_validator.py

leo\runtime\ingestion_legacy_v01_ingestion\truth_ingestor.py

leo\runtime\epistemic\ingestion.py

leo\runtime\epistemic\epistemic_audit_logger.py

leo\runtime\epistemic\v0_legacy\truth_ingestion_engine_v0_2.py

leo\runtime\epistemic\ingestion_v1\cto_model.py

leo\runtime\epistemic\ingestion_v1\validation.py

leo\runtime\epistemic\ingestion_v1\truth_ingestion_engine_v1.py

leo\runtime\epistemic\ingestion_v1\epistemic_memory_v1.py

leo\runtime\epistemic\ingestion_v1\test_ingestion_v1.py

leo\runtime\epistemic\ingestion_v1\external_verifier_v1.py

leo\runtime\epistemic\ingestion_v1\root_signing_v1.py

leo\runtime\epistemic\network_v1\consensus_engine_v1.py

leo\runtime\epistemic\network_v1\snapshot_verifier_v1.py

leo\runtime\epistemic\network_v1\trusted_registry_v1.py

leo\runtime\epistemic\network_v1\network_simulator_v1.py

leo\runtime\epistemic\network_v1\network_state_manager_v1.py

leo\runtime\epistemic\network_v1\divergence_analyzer_v1.py

leo\runtime\epistemic\network_v1\node_status_registry_v1.py

leo\runtime\epistemic\network_v1\state_root_broadcast.py

leo\runtime\epistemic\network_v1\state_root_consensus.py

leo\runtime\epistemic\recovery_v1\recovery_event_commit.py

leo\runtime\epistemic\recovery_v1\recovery_verifier.py

leo\runtime\epistemic\network_rebuild_v1\state_reconstructor.py

leo\runtime\epistemic\network_rebuild_v1\state_verifier.py

leo\runtime\epistemic\network_v2\core_registry.py

leo\runtime\epistemic\network_v2\core_handshake.py

leo\runtime\epistemic\network_v2\consensus_engine.py

leo\runtime\epistemic\network_v2\fork_detector.py

leo\runtime\epistemic\network_v2\state_canonizer.py

leo\runtime\epistemic\network_v2\consensus_logger.py

leo\runtime\epistemic\network_v2\octopus_network_simulator.py

leo\runtime\epistemic\network_v2\core_identity_generator.py

leo\runtime\epistemic\graph\entity_registry.py

leo\runtime\epistemic\graph__init__.py

leo\runtime\epistemic\graph\relation_registry.py

leo\runtime\epistemic\graph\graph_store.py

leo\runtime\epistemic\graph\graph_integrity.py

leo\runtime\epistemic\graph\epistemic_graph_engine.py

leo\runtime\epistemic\graph\graph_queries.py

leo\runtime\routing\core_definition.py

leo\runtime\routing\core_registry.py

leo\runtime\routing\core_router.py

leo\runtime\routing\core_topology_manager.py

leo\runtime\routing\cross_anchor_layer.py

leo\runtime\routing\topology_snapshot_commit.py

leo\runtime\routing\reconstruction_protocol.py

leo\runtime\routing\anchor_recovery_protocol.py

leo\runtime\identity\core_identity.py

leo\runtime\identity\identity_registry.py

leo\runtime\identity__init__.py

leo\runtime\identity\identity_signer.py

leo\runtime\identity\identity_storage.py

leo\runtime\domain_cores\domain_core_registry.py

leo\runtime\domain_cores\domain_interaction_engine.py

leo\runtime\domain_cores\systemic_risk_engine.py

leo\runtime\domain_cores\epistemic_core\epistemic_event_types.py

leo\runtime\domain_cores\epistemic_core\epistemic_event_classifier.py

leo\runtime\domain_cores\epistemic_core\epistemic_event_ingestion.py

leo\runtime\domain_cores\epistemic_core\epistemic_state_commit.py

leo\runtime\domain_cores\epistemic_core\epistemic_anomaly_engine.py

leo\runtime\domain_cores\epistemic_core\epistemic_contradiction_engine.py

leo\runtime\domain_cores\epistemic_core\epistemic_signal_engine.py

leo\runtime\domain_cores\epistemic_core\epistemic_truth_pipeline.py

leo\runtime\domain_cores\epistemic_core\epistemic_response_engine.py

leo\runtime\domain_cores\epistemic_core\epistemic_cross_domain_engine.py

leo\runtime\network\core_synchronization_engine.py

leo\runtime\alerts\institutional_alert_engine.py

leo\runtime\alerts\institution_routing_engine.py

leo\runtime\alerts\institution_response_tracker.py

leo\runtime\alerts\resolution_engine.py

leo\runtime\cases\case_engine.py

leo\runtime\memory\archive_manager.py

leo\runtime\system\leo_system_chronicle.py

leo\runtime\analysis\anomaly_engine.py

leo\runtime\analysis\contradiction_engine.py

leo\runtime\analysis\signal_engine.py

leo\runtime\analysis\epistemic_classifier.py

leo\runtime\analysis\anomaly_memory.py

leo\runtime\analysis\anomaly_pattern_engine.py

TOTAL MODULES: 119

CLASSES DETECTED:

AccountingEntry

AdmissionDecision

AdmissionResult

AdmissionStatus

AnchorRecoveryError

AnchorRecoveryProtocol

AnomalyEngine

AnomalyMemory

AnomalyPatternEngine

ArchiveManager

AuditVerificationError

BootIntegrityError

CanonicalTruthObject

CaseEngine

ConsensusEngineV1

ConsensusVerdict

ContradictionEngine

CoreDefinition

CoreIdentity

CoreRegistry

CoreRegistryError

CoreRouter

CoreRouterError

CoreStatus

CoreSynchronizationEngine

CoreTopologyError

CoreTopologyManager

CoreType

CrossAnchorError

CrossAnchorLayer

CrystalEnvelope

Decision

DivergenceAnalyzerV1

DivergenceResult

DomainCoreRegistry

DomainInteractionEngine

Engine

Entity

EntityRegistry

EpistemicAnomalyEngine

EpistemicAuditLogger

EpistemicBootIntegrityError

EpistemicCertificate

EpistemicClassifier

EpistemicContradictionEngine

EpistemicCrossDomainEngine

EpistemicDecision

EpistemicEvent

EpistemicEventClassifier

EpistemicEventIngestion

EpistemicEventType

EpistemicGatekeeper

EpistemicGraphEngine

EpistemicMemory

EpistemicMemoryError

EpistemicMemoryV1

EpistemicMemoryV1Error

EpistemicResponseEngine

EpistemicSignalEngine

EpistemicStateCommit

EpistemicTruthPipeline

ExecutionEpoch

GovernanceEntry

GovernanceSignal

GraphQueries

GraphStore

IdentityRegistry

IdentitySigner

IdentityStorage

IngestionService

InstitutionResponseTracker

InstitutionRoutingEngine

InstitutionalAlertEngine

IntegrityDebt

IntegrityDebtStatus

IntegrityError

Intent

LeoRuntimeV01

LeoSystemChronicle

NetworkMode

NetworkStateManagerV1

NodeStatus

NodeStatusRegistryV1

PersistedDebt

PersistedState

PersistedTruth

PersistenceCoordinator

ReconstructionError

ReconstructionProtocol

RelationRegistry

ReplayConsistencyError

ResolutionEngine

RootSignerV1

RuntimeContext

RuntimeFlow

SignalEngine

SnapshotVerdict

SnapshotVerifierV1

SourceType

StateCommit

StateHasher

StateRootBroadcast

StateRootConsensus

StateWriter

SystemBootCheck

SystemicRiskEngine

TamperAlert

TamperCheckResult

TopologySnapshotCommit

TopologySnapshotCommitError

TrustedKeyRegistryV1

TruthClassification

TruthIngestionEngine

TruthIngestionEngineV1

TruthIngestor

TruthObject

TruthSource

TruthState

TruthValidationError

TruthValue

ValidationError

TOTAL CLASSES: 121

FUNCTIONS DETECTED:

__init__

_apply_event

_build_merkle_proof

_build_merkle_root

_build_merkle_tree

_canonicalize

_cert

_check_execution_contract

_check_no_python_in_leo_core

_check_single_runtime_entry

_check_state_directory

_collect_structural_confirmations

_compute_hash

_compute_state_anchor

_create_certificate

_decision

_derive_certificate_id

_derive_truth_id

_determine_type

_evaluate

_get_last_state_hash

_get_last_truth_hash

_get_latest_topology_snapshot

_handle_confirmed

_handle_refused

_handle_silence

_hash

_hash_event

_hash_file

_hash_payload

_load

_load_events

_load_index

_load_or_create_identity

_load_state

_load_state_index

_log_escalation

_log_violation

_minimal_schema_valid

_next_version

_now

_on_confirmed

_on_refused

_on_silence

_persist

_persist_state

_persist_state_root_snapshot

_register_debt

_reject

_save

_save_key

_sign_payload

_stable_hash

_update_index

_validate_schema

activate_majority

add_relation

analyze

analyze_interaction

anomaly_statistics

append_certificate

append_edge

append_truth

archive_file

atomic_commit

boot_integrity_check

broadcast

broadcast_state

build_audit_record

build_cto

can_add_structural_core

canonize_state

check_resolution

classify

classify_truth

commit

commit_state

compare_states

compute_anchor_root

compute_global_state_root

compute_hash

compute_merkle_root

compute_state_anchor

consensus_on_state

create_anchor

create_case

create_entity

create_node

create_structural_core

decide

detect

detect_fork

detect_patterns

dispatch_latest_for_source

enforce_boot_integrity

ensure_node

evaluate

evaluate_snapshots

execute

find_relations_from

find_relations_to

fingerprint

from_certificate

from_private_key

full_integrity_scan

generate

generate_alert

generate_core_identity

generate_id

generate_key_pair

generate_merkle_proof

get

get_active_debts

get_all_edges

get_all_truths

get_anchor_core

get_anchor_root

get_core

get_fingerprint

get_identity

get_latest_snapshot

get_memory

get_mode

get_network_state

get_public_key

get_status

get_structural_cores

get_structural_snapshot

get_truth_engine

handle_intent

handle_red_incident

handshake

has_core

hash_edge

hash_payload

ingest

invariant_check

is_anchor

is_compromised

is_structural

is_trusted

is_valid

isolate_node

iter_audit_records

list_cores

list_entities

list_nodes

load_anomalies

load_or_create_anchor_key

load_recent

load_registry

load_secret_key

log

log_consensus_event

log_escalation

main

make_decision

mark_compromised

mark_trusted

normalize

now_utc

on_governance_signal

process_event

process_truth

read_all_edges

reconstruct

reconstruct_state_from_network

record_anomaly

record_event

record_recovery_event

record_response

recover

register

register_core

register_node

register_structural_root

register_truth

remaining_structural_slots

replay_consistency_gate

replay_from_audit

resolve

route

run_all_checks

run_tamper_checks

run_test

save_registry

set_mode

set_status

sign_payload

sign_root

sign_snapshot

simulate

simulate_network

start_new

to_dict

topology_snapshot

update_case_status

update_core_state

validate

validate_chain

validate_intent

validate_payload

validate_state_evolution

validate_state_schema

validate_topology

validate_truth

verify_anchor_consistency

verify_audit_log

verify_merkle_proof

verify_reconstructed_state

verify_record

```
verify_recovery_events
verify_root
verify_snapshot
verify_snapshot_signature
write_audit_log
write_state

TOTAL FUNCTIONS: 217

STORAGE CHECK:

epistemic_v1_storage  ✓ FOUND
identity_storage  ✓ FOUND
snapshot_storage  ✗ NOT FOUND
anomaly_storage  ✗ NOT FOUND

RUNTIME AUDIT COMPLETE

PS D:\BBS-09-01-2026> python -m tests_runtime.storage_integrity_audit

LEO STORAGE INTEGRITY AUDIT

FILES IN STORAGE:

  truths
  index
  state_roots
  identity
  epistemic_snapshot.json
  recovery_events.jsonl
  epistemic_events.jsonl
  epistemic_analysis_log.jsonl
  institutional_alerts.jsonl
  institution_responses.jsonl
  cases_registry.jsonl
  leo_system_chronicle.jsonl
  anomaly_library.jsonl
  epistemic_graph_edges.jsonl
```

```
--- FILE ANALYSIS ---


Analyzing: recovery_events.jsonl
 records: 1
 first hash: 42243f64d10c14ddbf03cd28b981759dbc945e8a69283fa91cadec3bdabf3f16
 last  hash: 42243f64d10c14ddbf03cd28b981759dbc945e8a69283fa91cadec3bdabf3f16

Analyzing: epistemic_events.jsonl
 records: 2
 first hash: d06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae
 last  hash: 3ad8cfab7cdb3284429c3bab729a6bce09e566f37c3d790b8266d960ce637bf6

Analyzing: epistemic_analysis_log.jsonl
 records: 1
 first hash: 9195240f800ad67fd08044a67f19691f66cf72a24f901b9ad121cd683f2269fe
 last  hash: 9195240f800ad67fd08044a67f19691f66cf72a24f901b9ad121cd683f2269fe

Analyzing: institutional_alerts.jsonl
 records: 0

Analyzing: institution_responses.jsonl
 records: 2
 first hash: aeb4d334224c973818b658acf7e63e68a7a7943ce25315d8dcc2bee79a61a0c5
 last  hash: 4e17095f756d4d9e4938d812d7d9c921bc5b2d534eace89926614573afd83d7c

Analyzing: cases_registry.jsonl
 records: 2
 first hash: 3989cf2fc5f14f72272ce0d318ab24340d3aa7d1cf5b658c6cd51c3c39905806
 last  hash: ecc4a43914274ca7ce5acade355526cc034cb6ea9ec4eb27f0012d0d351288cd

Analyzing: leo_system_chronicle.jsonl
 records: 1
 first hash: aa78804e30c80790ca72b6a91d548db315793ad3f589b24c557e20ab0ce90db7
 last  hash: aa78804e30c80790ca72b6a91d548db315793ad3f589b24c557e20ab0ce90db7

Analyzing: anomaly_library.jsonl
 records: 3
 first hash: d566cc141d0fceb4497f224bb7604e3e07e2fc11f173e55ea266ac215f8cb02f
```

```
 last  hash: 0b5286e3ed5045df1fa58549407b213c91d303b1a294786e446826df0973781f


Analyzing: epistemic_graph_edges.jsonl
 records: 1
 first hash: 3c90e392e3eec4056aa0726a72c5958ad2c1092c6723239d3099331ee4bd9749
 last  hash: 3c90e392e3eec4056aa0726a72c5958ad2c1092c6723239d3099331ee4bd9749


--- RECORD TYPES ---


ANOMALY: 3


TOTAL RECORDS: 13


STORAGE AUDIT COMPLETE


PS D:\BBS-09-01-2026> python -m tests_runtime.test_epistemic_graph
{'status': 'EDGE_RECORDED', 'edge_hash':
'998ddec0ad2ef7225fba0ba9e2508e073efa1c67e2fa8748c669f8f22f502bfb'}
{'status': 'DUPLICATE_EDGE_BLOCKED', 'edge': {'subject':
'Company:226450fc90b54102a5ddc56e898f4473', 'relation': 'OWNS', 'object':
'Company:e2d60a73335945d48358211c4abea1a5', 'timestamp': '2026-03-08T10:06:55.813476',
'source_fact_hash': None}}
GRAPH ROOT: 94edac9e2d00c78889edb2710c946056b061bae64ad71d2413cb61bbb5ac5e78
PS D:\BBS-09-01-2026> python -m tests_runtime.test_graph_propagation
PROPAGATION RESULT:
{'entity': 'Company:88c7b0872573445c9ee00d7c9a4a8db3', 'depth': 0}
{'entity': 'Company:50e9b1b7a75f418da7a7515bcd99b9aa', 'depth': 1}
{'entity': 'Company:77bbd0352e8c4c7ebfc1f657e4211ffd', 'depth': 2}
{'entity': 'Company:3e2f2429e2dc45af9aa6a844c559ba2e', 'depth': 3}
PS D:\BBS-09-01-2026> python -m tests_runtime.test_risk_engine
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "D:\BBS-09-01-2026\tests_runtime\test_risk_engine.py", line 7, in <module>
    report = engine.risk_report(entity)
  File "D:\BBS-09-01-2026\leo\runtime\analysis\epistemic_risk_engine.py", line 44, in risk_report
    "entity_risk": self.entity_risk_score(entity_id),
                   ~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^
  File "D:\BBS-09-01-2026\leo\runtime\analysis\epistemic_risk_engine.py", line 14, in
entity_risk_score
    anomalies = self.anomaly_memory.load_anomalies()
```

```
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
AttributeError: 'AnomalyMemory' object has no attribute 'load_anomalies'
PS D:\BBS-09-01-2026> python -m tests_runtime.test_risk_engine
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "D:\BBS-09-01-2026\tests_runtime\test_risk_engine.py", line 1, in <module>
    from leo.runtime.analysis.epistemic_risk_engine import EpistemicRiskEngine
  File "D:\BBS-09-01-2026\leo\runtime\analysis\epistemic_risk_engine.py", line 70
    }
    ^
SyntaxError: unmatched '}'
PS D:\BBS-09-01-2026> python -m tests_runtime.test_risk_engine
RISK REPORT:
{'entity': 'Company:test', 'entity_risk': 0, 'network_risk': 0}
PS D:\BBS-09-01-2026> python -m tests_runtime.test_pattern_engine
PATTERN DETECTION RESULT:
PS D:\BBS-09-01-2026> python leo_status.py

==============================
      LEO SYSTEM STATUS
==============================

Epistemic Memory
------------------------------
truth records: 2
events: 2

Graph
------------------------------
graph edges: 5
graph root: e54d46d7e4bc8e0d3b38e18e82e5939dae6b60033d8a39bff60d2359e6d4baff

Analysis
------------------------------
anomalies: 3

Governance
------------------------------
```

```
cases: 2

alerts: 0

responses: 2


Recovery / System

------------------------------

recovery events: 1

chronicle records: 1


==============================


PS D:\BBS-09-01-2026> python -m tests_runtime.test_epistemic_graph

{'status': 'EDGE_RECORDED', 'edge_hash':
'c2d251cbb2d4d311d6f74e39dfff8bd80b9169f0bd4744624976aa2e24adeaaa'}

{'status': 'DUPLICATE_EDGE_BLOCKED', 'edge': {'subject':
'Company:ff226c1c5cf54bbba11288548a38f1f4', 'relation': 'OWNS', 'object':
'Company:c03d0f8698c4445f987ee18c670c92db', 'event_time': '2026-03-08T11:07:17.817496',
'system_time': '2026-03-08T11:07:17.817496', 'timestamp': '2026-03-08T11:07:17.817496',
'source_fact_hash': None}}

GRAPH ROOT: e911b5269340a3391f9098abe4aa32714b17fdfdddff4598d74e026f12f8c8bf

PS D:\BBS-09-01-2026> python -m tests_runtime.test_graph_propagation

PROPAGATION RESULT:

{'entity': 'Company:7b9b356eef7f4bd8b918d78a8dbb2f9e', 'depth': 0}

{'entity': 'Company:20d4e0a5e46b495bb8f84438879af3c0', 'depth': 1}

{'entity': 'Company:2ab2c848bbb14c4ca607dd2f6408b534', 'depth': 2}

{'entity': 'Company:9169d2bc498747dbbfd05dd59cdefbad', 'depth': 3}

PS D:\BBS-09-01-2026> python -m tests_runtime.test_risk_engine

RISK REPORT:

{'entity': 'Company:test', 'entity_risk': 0, 'network_risk': 0}

PS D:\BBS-09-01-2026> python leo_status.py


==============================

      LEO SYSTEM STATUS

==============================


Epistemic Memory

------------------------------

truth records: 2

events: 2


Graph
```

```
------------------------------

graph edges: 9

graph root: 073d860eb07e4babf0e61bb70047e0ed447f92e58c5be63f7b043b73f50691aa


Analysis

------------------------------

anomalies: 3


Governance

------------------------------

cases: 2

alerts: 0

responses: 2


Recovery / System

------------------------------

recovery events: 1

chronicle records: 1


==============================


PS D:\BBS-09-01-2026> python -m tests_runtime.test_temporal_anomaly

TEMPORAL ANOMALIES:

{'type': 'RAPID_CHAIN', 'relation': 'OWNS', 'first': {'subject':
'Company:ff226c1c5cf54bbba11288548a38f1f4', 'object': 'Company:c03d0f8698c4445f987ee18c670c92db',
'time': datetime.datetime(2026, 3, 8, 11, 7, 17, 810449)}, 'second': {'subject':
'Company:7b9b356eef7f4bd8b918d78a8dbb2f9e', 'object': 'Company:20d4e0a5e46b495bb8f84438879af3c0',
'time': datetime.datetime(2026, 3, 8, 11, 8, 16, 147547)}, 'hours_between': 0.016204749444444444}

{'type': 'RAPID_CHAIN', 'relation': 'OWNS', 'first': {'subject':
'Company:7b9b356eef7f4bd8b918d78a8dbb2f9e', 'object': 'Company:20d4e0a5e46b495bb8f84438879af3c0',
'time': datetime.datetime(2026, 3, 8, 11, 8, 16, 147547)}, 'second': {'subject':
'Company:20d4e0a5e46b495bb8f84438879af3c0', 'object': 'Company:2ab2c848bbb14c4ca607dd2f6408b534',
'time': datetime.datetime(2026, 3, 8, 11, 8, 16, 155484)}, 'hours_between': 2.2047222222222222e-
06}

{'type': 'RAPID_CHAIN', 'relation': 'OWNS', 'first': {'subject':
'Company:20d4e0a5e46b495bb8f84438879af3c0', 'object': 'Company:2ab2c848bbb14c4ca607dd2f6408b534',
'time': datetime.datetime(2026, 3, 8, 11, 8, 16, 155484)}, 'second': {'subject':
'Company:2ab2c848bbb14c4ca607dd2f6408b534', 'object': 'Company:9169d2bc498747dbbfd05dd59cdefbad',
'time': datetime.datetime(2026, 3, 8, 11, 8, 16, 162434)}, 'hours_between': 1.9305555555555554e-
06}

PS D:\BBS-09-01-2026> python -m tests_runtime.test_case_evidence

CASE EVIDENCE:


ENTITY:

Company:test
```

RELATIONS:

TIMELINE:

TEMPORAL ANOMALIES:

{'type': 'RAPID_CHAIN', 'relation': 'OWNS', 'first': {'subject': 'Company:ff226c1c5cf54bbba11288548a38f1f4', 'object': 'Company:c03d0f8698c4445f987ee18c670c92db', 'time': datetime.datetime(2026, 3, 8, 11, 7, 17, 810449)}, 'second': {'subject': 'Company:7b9b356eef7f4bd8b918d78a8dbb2f9e', 'object': 'Company:20d4e0a5e46b495bb8f84438879af3c0', 'time': datetime.datetime(2026, 3, 8, 11, 8, 16, 147547)}, 'hours_between': 0.016204749444444444}

{'type': 'RAPID_CHAIN', 'relation': 'OWNS', 'first': {'subject': 'Company:7b9b356eef7f4bd8b918d78a8dbb2f9e', 'object': 'Company:20d4e0a5e46b495bb8f84438879af3c0', 'time': datetime.datetime(2026, 3, 8, 11, 8, 16, 147547)}, 'second': {'subject': 'Company:20d4e0a5e46b495bb8f84438879af3c0', 'object': 'Company:2ab2c848bbb14c4ca607dd2f6408b534', 'time': datetime.datetime(2026, 3, 8, 11, 8, 16, 155484)}, 'hours_between': 2.2047222222222222e-06}

{'type': 'RAPID_CHAIN', 'relation': 'OWNS', 'first': {'subject': 'Company:20d4e0a5e46b495bb8f84438879af3c0', 'object': 'Company:2ab2c848bbb14c4ca607dd2f6408b534', 'time': datetime.datetime(2026, 3, 8, 11, 8, 16, 155484)}, 'second': {'subject': 'Company:2ab2c848bbb14c4ca607dd2f6408b534', 'object': 'Company:9169d2bc498747dbbfd05dd59cdefbad', 'time': datetime.datetime(2026, 3, 8, 11, 8, 16, 162434)}, 'hours_between': 1.9305555555555554e-06}

RISK:

{'entity': 'Company:test', 'entity_risk': 0, 'network_risk': 0}

```
PS D:\BBS-09-01-2026> python leo_system_map.py


==============================

        LEO SYSTEM MAP

==============================


ENTITIES: 14

  Company:20d4e0a5e46b495bb8f84438879af3c0

  Company:226450fc90b54102a5ddc56e898f4473

  Company:2ab2c848bbb14c4ca607dd2f6408b534

  Company:3e2f2429e2dc45af9aa6a844c559ba2e

  Company:50e9b1b7a75f418da7a7515bcd99b9aa

  Company:7256eabaae37459e9c780b0af7fd2c61

  Company:77bbd0352e8c4c7ebfc1f657e4211ffd

  Company:7b9b356eef7f4bd8b918d78a8dbb2f9e

  Company:88c7b0872573445c9ee00d7c9a4a8db3

  Company:9169d2bc498747dbbfd05dd59cdefbad

  Company:c03d0f8698c4445f987ee18c670c92db

  Company:e2d60a73335945d48358211c4abea1a5
```

```
  Company:f89765700d4f42d2932c2cdfe363e4bb

  Company:ff226c1c5cf54bbba11288548a38f1f4


RELATIONS: 9

Company:f89765700d4f42d2932c2cdfe363e4bb OWNS Company:7256eabaae37459e9c780b0af7fd2c61

Company:226450fc90b54102a5ddc56e898f4473 OWNS Company:e2d60a73335945d48358211c4abea1a5

Company:88c7b0872573445c9ee00d7c9a4a8db3 OWNS Company:50e9b1b7a75f418da7a7515bcd99b9aa

Company:50e9b1b7a75f418da7a7515bcd99b9aa OWNS Company:77bbd0352e8c4c7ebfc1f657e4211ffd

Company:77bbd0352e8c4c7ebfc1f657e4211ffd OWNS Company:3e2f2429e2dc45af9aa6a844c559ba2e

Company:ff226c1c5cf54bbba11288548a38f1f4 OWNS Company:c03d0f8698c4445f987ee18c670c92db

Company:7b9b356eef7f4bd8b918d78a8dbb2f9e OWNS Company:20d4e0a5e46b495bb8f84438879af3c0

Company:20d4e0a5e46b495bb8f84438879af3c0 OWNS Company:2ab2c848bbb14c4ca607dd2f6408b534

Company:2ab2c848bbb14c4ca607dd2f6408b534 OWNS Company:9169d2bc498747dbbfd05dd59cdefbad


ANOMALIES: 3

{'record_type': 'ANOMALY', 'timestamp': '2026-03-08T08:20:06.346267', 'anomaly': {'type':
'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'deviation': 2.7, 'contract_id': 'PROC-
77821'}}

{'record_type': 'ANOMALY', 'timestamp': '2026-03-08T08:46:38.835975', 'anomaly': {'type':
'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'deviation': 2.7, 'contract_id': 'PROC-
77821'}}

{'record_type': 'ANOMALY', 'timestamp': '2026-03-08T08:47:13.254040', 'anomaly': {'type':
'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'deviation': 2.7, 'contract_id': 'PROC-
77821'}}


CASES: 2

{'timestamp': '2026-03-07T17:40:47.262182', 'case_id': 'CASE_4ac321f3a93a', 'alert_id':
'ALERT_001', 'domains': ['finance', 'medicine'], 'severity': 'HIGH', 'status':
'UNDER_INVESTIGATION', 'updated_at': '2026-03-07T17:40:47.278475'}

{'timestamp': '2026-03-07T19:11:50.170965', 'case_id': 'CASE_b4f0c0e58c8a', 'alert_id':
'ALERT_001', 'domains': ['finance'], 'severity': 'HIGH', 'status': 'OPEN'}


ALERTS: 0


==============================


PS D:\BBS-09-01-2026> python -m tests_runtime.test_case_trigger

CASE TRIGGER RESULT:


{'status': 'CASE_TRIGGERED', 'case': {'status': 'CASE_CREATED', 'case_id': 'CASE_f250d292d1b3'},
'entity': 'Company:20d4e0a5e46b495bb8f84438879af3c0'}

PS D:\BBS-09-01-2026> python -m tests_runtime.test_institutional_alert_engine

{'status': 'INSTITUTIONAL_ALERT_CREATED', 'alert_level': 'CRITICAL', 'file':
'epistemic_v1_storage\\institutional_alerts.jsonl'}
```

```
PS D:\BBS-09-01-2026> python -m tests_runtime.test_pattern_library

CORRUPTION PATTERN LIBRARY:


SHELL_CHAIN

{'description': 'Rapid multi-level ownership chain', 'relations': ['OWNS', 'OWNS', 'OWNS'],
'max_time_hours': 24}


OWNERSHIP_BURST

{'description': 'Multiple ownership relations created in short time', 'relations': ['OWNS'],
'burst_threshold': 5, 'time_window_hours': 1}


CONTRACT_CAPTURE

{'description': 'Entity wins contract shortly after creation', 'relations': ['CREATED',
'WINS_CONTRACT'], 'max_time_hours': 48}


SUPPLIER_CARTEL

{'description': 'Multiple suppliers linked through shared ownership', 'relations': ['OWNS',
'SUPPLIES'], 'cluster_size': 3}


MONEY_LOOP

{'description': 'Circular transfer chain', 'relations': ['TRANSFER', 'TRANSFER', 'TRANSFER'],
'loop_required': True}


PS D:\BBS-09-01-2026> python -m tests_runtime.test_pattern_detection

Traceback (most recent call last):

  File "<frozen runpy>", line 198, in _run_module_as_main

  File "<frozen runpy>", line 88, in _run_code

  File "D:\BBS-09-01-2026\tests_runtime\test_pattern_detection.py", line 9, in <module>

    patterns = engine.detect_patterns(edges)

               ^^^^^^^^^^^^^^^^^^^^^^^

AttributeError: 'AnomalyPatternEngine' object has no attribute 'detect_patterns'. Did you mean:
'detect_graph_patterns'?

PS D:\BBS-09-01-2026> python -m tests_runtime.test_pattern_detection

PATTERN DETECTION RESULT:


anomaly_patterns

graph_patterns

PS D:\BBS-09-01-2026> python -m tests_runtime.test_pattern_detection

PATTERN DETECTION RESULT:


ANOMALY PATTERNS:
```

```
{'patterns_detected': 1, 'patterns': [{'pattern_type': 'PRICE_ANOMALY', 'domain': 'finance',
'severity': 'HIGH', 'occurrences': 3}]}


GRAPH PATTERNS:

[{'pattern': 'SHELL_CHAIN', 'description': 'Rapid multi-level ownership chain'}, {'pattern':
'OWNERSHIP_BURST', 'description': 'Multiple ownership relations created in short time'}]

PS D:\BBS-09-01-2026> python -m tests_runtime.test_graph_influence_engine

GRAPH INFLUENCE RANKING:


{'entity': 'Company:50e9b1b7a75f418da7a7515bcd99b9aa', 'influence_score': 2}

{'entity': 'Company:77bbd0352e8c4c7ebfc1f657e4211ffd', 'influence_score': 2}

{'entity': 'Company:20d4e0a5e46b495bb8f84438879af3c0', 'influence_score': 2}

{'entity': 'Company:2ab2c848bbb14c4ca607dd2f6408b534', 'influence_score': 2}

{'entity': 'Company:f89765700d4f42d2932c2cdfe363e4bb', 'influence_score': 1}

{'entity': 'Company:7256eabaae37459e9c780b0af7fd2c61', 'influence_score': 1}

{'entity': 'Company:226450fc90b54102a5ddc56e898f4473', 'influence_score': 1}

{'entity': 'Company:e2d60a73335945d48358211c4abea1a5', 'influence_score': 1}

{'entity': 'Company:88c7b0872573445c9ee00d7c9a4a8db3', 'influence_score': 1}

{'entity': 'Company:3e2f2429e2dc45af9aa6a844c559ba2e', 'influence_score': 1}

PS D:\BBS-09-01-2026> python -m tests_runtime.test_corruption_cluster_engine

C:\Users\user\AppData\Local\Programs\Python\Python313\python.exe: No module named
tests_runtime.test_corruption_cluster_engine

PS D:\BBS-09-01-2026> python -m tests_runtime.test_corruption_cluster_engine

CORRUPTION CLUSTERS:


{'cluster_size': 4, 'entities': ['Company:3e2f2429e2dc45af9aa6a844c559ba2e',
'Company:88c7b0872573445c9ee00d7c9a4a8db3', 'Company:77bbd0352e8c4c7ebfc1f657e4211ffd',
'Company:50e9b1b7a75f418da7a7515bcd99b9aa']}

{'cluster_size': 4, 'entities': ['Company:7b9b356eef7f4bd8b918d78a8dbb2f9e',
'Company:2ab2c848bbb14c4ca607dd2f6408b534', 'Company:9169d2bc498747dbbfd05dd59cdefbad',
'Company:20d4e0a5e46b495bb8f84438879af3c0']}

PS D:\BBS-09-01-2026> python -m tests_runtime.test_evidence_graph_builder

Traceback (most recent call last):

  File "<frozen runpy>", line 198, in _run_module_as_main

  File "<frozen runpy>", line 88, in _run_code

  File "D:\BBS-09-01-2026\tests_runtime\test_evidence_graph_builder.py", line 7, in <module>

    evidence = builder.build_evidence_package(entity)

  File "D:\BBS-09-01-2026\leo\runtime\analysis\evidence_graph_builder.py", line 41, in
build_evidence_package

    anomalies = self.anomaly_memory.load_anomalies()

                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

AttributeError: 'AnomalyMemory' object has no attribute 'load_anomalies'
```

```
PS D:\BBS-09-01-2026> python -m tests_runtime.test_evidence_graph_builder
```

EVIDENCE PACKAGE

ENTITY

Company:test

RELATIONS

[{'edge_hash': 'fd8aa03cb9947d9cd480958a65dad2c6311b00d371d463ee50b9872a4eb38e4c', 'edge': {'subject': 'Company:f89765700d4f42d2932c2cdfe363e4bb', 'relation': 'OWNS', 'object': 'Company:7256eabaae37459e9c780b0af7fd2c61', 'timestamp': '2026-03-08T09:26:43.859328', 'source_fact_hash': None}}, {'edge_hash': '998ddec0ad2ef7225fba0ba9e2508e073efa1c67e2fa8748c669f8f22f502bfb', 'edge': {'subject': 'Company:226450fc90b54102a5ddc56e898f4473', 'relation': 'OWNS', 'object': 'Company:e2d60a73335945d48358211c4abea1a5', 'timestamp': '2026-03-08T10:06:55.806793', 'source_fact_hash': None}}, {'edge_hash': 'a8e9209651bbce4a4894090a85c76e572bd77380d6879f7f9b09681af5cbbe4f', 'edge': {'subject': 'Company:88c7b0872573445c9ee00d7c9a4a8db3', 'relation': 'OWNS', 'object': 'Company:50e9b1b7a75f418da7a7515bcd99b9aa', 'timestamp': '2026-03-08T10:12:14.098333', 'source_fact_hash': None}}, {'edge_hash': 'c2e2d8e25ce7bde6c2584727478677214527 3f013e58ec7dc206d9eab22d7e8d', 'edge': {'subject': 'Company:50e9b1b7a75f418da7a7515bcd99b9aa', 'relation': 'OWNS', 'object': 'Company:77bbd0352e8c4c7ebfc1f657e4211ffd', 'timestamp': '2026-03-08T10:12:14.105326', 'source_fact_hash': None}}, {'edge_hash': '535ec1f59fd328cf940f981851412a143ed50231a3ebae3b4143982a3532ecfe', 'edge': {'subject': 'Company:77bbd0352e8c4c7ebfc1f657e4211ffd', 'relation': 'OWNS', 'object': 'Company:3e2f2429e2dc45af9aa6a844c559ba2e', 'timestamp': '2026-03-08T10:12:14.112251', 'source_fact_hash': None}}, {'edge_hash': 'c2d251cbb2d4d311d6f74e39dfff8bd80b9169f0bd4744624976aa2e24adeaaa', 'edge': {'subject': 'Company:ff226c1c5cf54bbba11288548a38f1f4', 'relation': 'OWNS', 'object': 'Company:c03d0f8698c4445f987ee18c670c92db', 'event_time': '2026-03-08T11:07:17.810449', 'system_time': '2026-03-08T11:07:17.810449', 'timestamp': '2026-03-08T11:07:17.810449', 'source_fact_hash': None}}, {'edge_hash': '78f36e924c8785db04c88b004c56ed91c2572c0e44c46fd5b3229a41cd58dcf7', 'edge': {'subject': 'Company:7b9b356eef7f4bd8b918d78a8dbb2f9e', 'relation': 'OWNS', 'object': 'Company:20d4e0a5e46b495bb8f84438879af3c0', 'event_time': '2026-03-08T11:08:16.147547', 'system_time': '2026-03-08T11:08:16.147547', 'timestamp': '2026-03-08T11:08:16.147547', 'source_fact_hash': None}}, {'edge_hash': 'f169e3052c45558bffe51482785f782acfdb038c6f75c26e304e5c1bd4af3164', 'edge': {'subject': 'Company:20d4e0a5e46b495bb8f84438879af3c0', 'relation': 'OWNS', 'object': 'Company:2ab2c848bbb14c4ca607dd2f6408b534', 'event_time': '2026-03-08T11:08:16.155484', 'system_time': '2026-03-08T11:08:16.155484', 'timestamp': '2026-03-08T11:08:16.155484', 'source_fact_hash': None}}, {'edge_hash': 'bd9fc5c65272be8aba2b717827ab1f70d7df3ce7165bc5a8c0b46d698f45d6b8', 'edge': {'subject': 'Company:2ab2c848bbb14c4ca607dd2f6408b534', 'relation': 'OWNS', 'object': 'Company:9169d2bc498747dbbfd05dd59cdefbad', 'event_time': '2026-03-08T11:08:16.162434', 'system_time': '2026-03-08T11:08:16.162434', 'timestamp': '2026-03-08T11:08:16.162434', 'source_fact_hash': None}}]

PATTERNS

{'anomaly_patterns': {'patterns_detected': 1, 'patterns': [{'pattern_type': 'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'occurrences': 3}]}, 'graph_patterns': [{'pattern': 'SHELL_CHAIN', 'description': 'Rapid multi-level ownership chain'}, {'pattern': 'OWNERSHIP_BURST', 'description': 'Multiple ownership relations created in short time'}]}

CLUSTERS

[{'cluster_size': 4, 'entities': ['Company:50e9b1b7a75f418da7a7515bcd99b9aa', 'Company:88c7b0872573445c9ee00d7c9a4a8db3', 'Company:77bbd0352e8c4c7ebfc1f657e4211ffd',

'Company:3e2f2429e2dc45af9aa6a844c559ba2e']}, {'cluster_size': 4, 'entities': ['Company:7b9b356eef7f4bd8b918d78a8dbb2f9e', 'Company:2ab2c848bbb14c4ca607dd2f6408b534', 'Company:9169d2bc498747dbbfd05dd59cdefbad', 'Company:20d4e0a5e46b495bb8f84438879af3c0']}]

INFLUENCE_RANKING

[{'entity': 'Company:50e9b1b7a75f418da7a7515bcd99b9aa', 'influence_score': 2}, {'entity': 'Company:77bbd0352e8c4c7ebfc1f657e4211ffd', 'influence_score': 2}, {'entity': 'Company:20d4e0a5e46b495bb8f84438879af3c0', 'influence_score': 2}, {'entity': 'Company:2ab2c848bbb14c4ca607dd2f6408b534', 'influence_score': 2}, {'entity': 'Company:f89765700d4f42d2932c2cdfe363e4bb', 'influence_score': 1}, {'entity': 'Company:7256eabaae37459e9c780b0af7fd2c61', 'influence_score': 1}, {'entity': 'Company:226450fc90b54102a5ddc56e898f4473', 'influence_score': 1}, {'entity': 'Company:e2d60a73335945d48358211c4abea1a5', 'influence_score': 1}, {'entity': 'Company:88c7b0872573445c9ee00d7c9a4a8db3', 'influence_score': 1}, {'entity': 'Company:3e2f2429e2dc45af9aa6a844c559ba2e', 'influence_score': 1}]

ANOMALIES

[{'type': 'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'deviation': 2.7, 'contract_id': 'PROC-77821'}, {'type': 'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'deviation': 2.7, 'contract_id': 'PROC-77821'}, {'type': 'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'deviation': 2.7, 'contract_id': 'PROC-77821'}]

```
PS D:\BBS-09-01-2026> python -m tests_runtime.test_investigation_report_generator
```

INVESTIGATION REPORT

REPORT_GENERATED_AT

2026-03-08T21:08:22.726443

ENTITY

Company:test

SUMMARY

Detected 1 repeating anomaly pattern(s). Detected 2 structural network pattern(s). Identified 2 connected entity cluster(s). 3 financial anomaly record(s) present in evidence.

NETWORK_SIZE

9

CLUSTERS_DETECTED

2

TOP_INFLUENTIAL_ENTITIES

[{'entity': 'Company:50e9b1b7a75f418da7a7515bcd99b9aa', 'influence_score': 2}, {'entity': 'Company:77bbd0352e8c4c7ebfc1f657e4211ffd', 'influence_score': 2}, {'entity': 'Company:20d4e0a5e46b495bb8f84438879af3c0', 'influence_score': 2}, {'entity': 'Company:2ab2c848bbb14c4ca607dd2f6408b534', 'influence_score': 2}, {'entity': 'Company:f89765700d4f42d2932c2cdfe363e4bb', 'influence_score': 1}]

DETECTED_PATTERNS

[{'pattern': 'SHELL_CHAIN', 'description': 'Rapid multi-level ownership chain'}, {'pattern': 'OWNERSHIP_BURST', 'description': 'Multiple ownership relations created in short time'}]

FINANCIAL_ANOMALIES

[{'type': 'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'deviation': 2.7, 'contract_id': 'PROC-77821'}, {'type': 'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'deviation': 2.7, 'contract_id': 'PROC-77821'}, {'type': 'PRICE_ANOMALY', 'domain': 'finance', 'severity': 'HIGH', 'deviation': 2.7, 'contract_id': 'PROC-77821'}]

NOTE

This report presents analytical signals and structured evidence. It does not determine guilt or legal responsibility.

PS D:\BBS-09-01-2026> python -m tests_runtime.test_investigation_timeline_engine

INVESTIGATION TIMELINE

{'time': '2026-03-08T09:26:43.859328', 'type': 'RELATION', 'description': 'Company:f89765700d4f42d2932c2cdfe363e4bb OWNS Company:7256eabaae37459e9c780b0af7fd2c61'}

{'time': '2026-03-08T10:06:55.806793', 'type': 'RELATION', 'description': 'Company:226450fc90b54102a5ddc56e898f4473 OWNS Company:e2d60a73335945d48358211c4abea1a5'}

{'time': '2026-03-08T10:12:14.098333', 'type': 'RELATION', 'description': 'Company:88c7b0872573445c9ee00d7c9a4a8db3 OWNS Company:50e9b1b7a75f418da7a7515bcd99b9aa'}

{'time': '2026-03-08T10:12:14.105326', 'type': 'RELATION', 'description': 'Company:50e9b1b7a75f418da7a7515bcd99b9aa OWNS Company:77bbd0352e8c4c7ebfc1f657e4211ffd'}

{'time': '2026-03-08T10:12:14.112251', 'type': 'RELATION', 'description': 'Company:77bbd0352e8c4c7ebfc1f657e4211ffd OWNS Company:3e2f2429e2dc45af9aa6a844c559ba2e'}

{'time': '2026-03-08T11:07:17.810449', 'type': 'RELATION', 'description': 'Company:ff226c1c5cf54bbba11288548a38f1f4 OWNS Company:c03d0f8698c4445f987ee18c670c92db'}

{'time': '2026-03-08T11:08:16.147547', 'type': 'RELATION', 'description': 'Company:7b9b356eef7f4bd8b918d78a8dbb2f9e OWNS Company:20d4e0a5e46b495bb8f84438879af3c0'}

{'time': '2026-03-08T11:08:16.155484', 'type': 'RELATION', 'description': 'Company:20d4e0a5e46b495bb8f84438879af3c0 OWNS Company:2ab2c848bbb14c4ca607dd2f6408b534'}

{'time': '2026-03-08T11:08:16.162434', 'type': 'RELATION', 'description': 'Company:2ab2c848bbb14c4ca607dd2f6408b534 OWNS Company:9169d2bc498747dbbfd05dd59cdefbad'}

{'time': '2026-03-09T16:16:04.185761', 'type': 'ANOMALY', 'description': 'PRICE_ANOMALY detected in domain finance'}

{'time': '2026-03-09T16:16:04.185779', 'type': 'ANOMALY', 'description': 'PRICE_ANOMALY detected in domain finance'}

{'time': '2026-03-09T16:16:04.185784', 'type': 'ANOMALY', 'description': 'PRICE_ANOMALY detected in domain finance'}

PS D:\BBS-09-01-2026> python -m tests_runtime.test_risk_escalation_engine

RISK ESCALATION RESULT

```
{'risk_level': 'CRITICAL', 'risk_score': 11, 'reasons': ['3 financial anomalies detected',
'repeating anomaly patterns detected', '2 structural graph patterns detected', 'cluster size 4
detected']}

PS D:\BBS-09-01-2026> python -m tests_runtime.test_case_escalation_engine

Traceback (most recent call last):

  File "<frozen runpy>", line 198, in _run_module_as_main

  File "<frozen runpy>", line 88, in _run_code

  File "D:\BBS-09-01-2026\tests_runtime\test_case_escalation_engine.py", line 15, in <module>

    result = escalation_engine.escalate(entity, risk)

  File "D:\BBS-09-01-2026\leo\runtime\analysis\case_escalation_engine.py", line 45, in escalate

    alert = self.alert_engine.create_alert(

            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

AttributeError: 'InstitutionalAlertEngine' object has no attribute 'create_alert'. Did you mean:
'generate_alert'?

PS D:\BBS-09-01-2026> python -m tests_runtime.test_case_escalation_engine

Traceback (most recent call last):

  File "<frozen runpy>", line 198, in _run_module_as_main

  File "<frozen runpy>", line 88, in _run_code

  File "D:\BBS-09-01-2026\tests_runtime\test_case_escalation_engine.py", line 15, in <module>

    result = escalation_engine.escalate(entity, risk)

  File "D:\BBS-09-01-2026\leo\runtime\analysis\case_escalation_engine.py", line 45, in escalate

    alert = self.alert_engine.generate_alert(

        alert_level="CRITICAL"

    )

TypeError: InstitutionalAlertEngine.generate_alert() got an unexpected keyword argument
'alert_level'

PS D:\BBS-09-01-2026> python -m tests_runtime.test_case_escalation_engine

CASE ESCALATION RESULT



{'action': 'CRITICAL_ESCALATION', 'alert': {'status': 'INSTITUTIONAL_ALERT_CREATED',
'alert_level': 'INFO', 'file': 'epistemic_v1_storage\\institutional_alerts.jsonl'}, 'case':
{'status': 'CASE_CREATED', 'case_id': 'CASE_b0ad4ac51cd6'}}

PS D:\BBS-09-01-2026>

Windows PowerShell

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Try the new cross-platform PowerShell https://aka.ms/pscore6



PS C:\Users\user> cd D:\BBS-09-01-2026

PS D:\BBS-09-01-2026> dir
```

```
    Directory: D:\BBS-09-01-2026


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        12.01.2026     20:34                leo-core
d-----        20.01.2026      7:47                modules
d-----        23.01.2026     18:10                foundation
d-----        02.02.2026      5:08                leo
d-----        14.02.2026     17:31                data
d-----        14.02.2026     17:31                archives
d-----        14.02.2026     17:37                sandbox
d-----        14.02.2026     17:48                tools
d-----        15.02.2026      9:22                .pytest_cache
d-----        22.02.2026     11:50                epistemic_v1_storage_mirror
d-----        22.02.2026     16:32                epistemic_v1_storage
d-----        02.03.2026      8:00                epistemic_network_v1
d-----        03.03.2026     20:03                __pycache__
d-----        04.03.2026     21:18                identity_storage
-a----        16.02.2026     18:33             31 payload.json
-a----        17.02.2026      7:53           5051 SYSTEM_INVENTORY_REPORT.json
-a----        17.02.2026      8:08          48182 leo_full_tree_snapshot.txt
-a----        20.02.2026     20:42          93560 SYSTEM_TREE_FULL.txt
-a----        20.02.2026     20:42         103815 FILE_INDEX.csv
-a----        02.03.2026      8:00         107282 SYSTEM_STRUCTURE_SNAPSHOT.txt
-a----        02.03.2026      8:00           4504 PYTHON_ENV_SNAPSHOT.txt
-a----        03.03.2026     20:39            417 manual_test.py
-a----        03.03.2026     22:06           1317 test_multicore.py
-a----        03.03.2026     22:11           1249 test_topology.py
-a----        03.03.2026     22:19           1287 test_cross_anchor.py
-a----        04.03.2026     20:02           1680 test_topology_commit.py
-a----        04.03.2026     20:25           1623 test_reconstruction.py
-a----        04.03.2026     21:36            721 test_anchor_recovery.py
-a----        05.03.2026     19:50            549 test_recovery_commit.py


PS D:\BBS-09-01-2026> dir leo\runtime\epistemic\recovery_v1
```

```
    Directory: D:\BBS-09-01-2026\leo\runtime\epistemic\recovery_v1



Mode                 LastWriteTime         Length Name

----                 -------------         ------ ----

-a----        05.03.2026     19:49           2353 recovery_event_commit.py



PS D:\BBS-09-01-2026> python test_recovery_commit.py

Traceback (most recent call last):

  File "D:\BBS-09-01-2026\test_recovery_commit.py", line 7, in <module>

    result = record_recovery_event(

        storage_path=storage,

    ...<3 lines>...

        anchor_private_key_path=anchor_key

    )

  File "D:\BBS-09-01-2026\leo\runtime\epistemic\recovery_v1\recovery_event_commit.py", line 66,
in record_recovery_event

    signature = _sign_payload(anchor_private_key_path, payload_hash)

  File "D:\BBS-09-01-2026\leo\runtime\epistemic\recovery_v1\recovery_event_commit.py", line 24,
in _sign_payload

    with open(private_key_path, "rb") as f:

         ~~~~^^^^^^^^^^^^^^^^^^^^^^^^^

FileNotFoundError: [Errno 2] No such file or directory: 'identity\\private_key.pem'

PS D:\BBS-09-01-2026> python test_recovery_commit.py

Traceback (most recent call last):

  File "D:\BBS-09-01-2026\test_recovery_commit.py", line 7, in <module>

    result = record_recovery_event(

        storage_path=storage,

    ...<3 lines>...

        anchor_private_key_path=anchor_key

    )

  File "D:\BBS-09-01-2026\leo\runtime\epistemic\recovery_v1\recovery_event_commit.py", line 66,
in record_recovery_event

    signature = _sign_payload(anchor_private_key_path, payload_hash)

  File "D:\BBS-09-01-2026\leo\runtime\epistemic\recovery_v1\recovery_event_commit.py", line 24,
in _sign_payload

    with open(private_key_path, "rb") as f:

         ~~~~^^^^^^^^^^^^^^^^^^^^^^^^^

FileNotFoundError: [Errno 2] No such file or directory: 'identity_storage\\private_key.pem'

PS D:\BBS-09-01-2026> python test_recovery_commit.py
```

```
Traceback (most recent call last):
  File "D:\BBS-09-01-2026\test_recovery_commit.py", line 7, in <module>
    result = record_recovery_event(
        storage_path=storage,
    ...<3 lines>...
        anchor_private_key_path=anchor_key
    )
  File "D:\BBS-09-01-2026\leo\runtime\epistemic\recovery_v1\recovery_event_commit.py", line 66, in record_recovery_event
    signature = _sign_payload(anchor_private_key_path, payload_hash)
  File "D:\BBS-09-01-2026\leo\runtime\epistemic\recovery_v1\recovery_event_commit.py", line 30, in _sign_payload
    signature = private_key.sign(
        payload_hash.encode(),
    ...<4 lines>...
        hashes.SHA256()
    )
TypeError: Ed25519PrivateKey.sign() takes 1 positional arguments but 3 were given
PS D:\BBS-09-01-2026> python test_recovery_commit.py
{'status': 'RECOVERY_EVENT_RECORDED', 'payload_hash': '16d7b37a368aa0e10f40625fd2223441fd42bcdb8a59ffd44fca481efcb4f873', 'storage_file': 'epistemic_v1_storage\\recovery_events.jsonl'}
PS D:\BBS-09-01-2026> python test_recovery_verifier.py
Traceback (most recent call last):
  File "D:\BBS-09-01-2026\test_recovery_verifier.py", line 7, in <module>
    result = verify_recovery_events(
        storage_path=storage,
        anchor_public_key_path=public_key
    )
  File "D:\BBS-09-01-2026\leo\runtime\epistemic\recovery_v1\recovery_verifier.py", line 29, in verify_recovery_events
    with open(anchor_public_key_path, "rb") as f:
         ~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
FileNotFoundError: [Errno 2] No such file or directory: 'identity_storage\\anchor_public_key.pem'
PS D:\BBS-09-01-2026> python export_anchor_public_key.py
ANCHOR PUBLIC KEY EXPORTED
identity_storage\anchor_public_key.pem
PS D:\BBS-09-01-2026> python test_recovery_verifier.py
{'status': 'VERIFICATION_COMPLETE', 'verified_events': 1, 'invalid_events': 0}
PS D:\BBS-09-01-2026> python test_node_rebuild.py
{'status': 'STATE_RECONSTRUCTED', 'events_recovered': 1}
```

```
PS D:\BBS-09-01-2026> python test_state_verifier.py

{'status': 'STATE_VERIFIED', 'events': 1, 'merkle_root':
'fd576b1cb55785524d0a8576b3badb4f72fa3091123bfbae2fbd68cada16aa11'}

PS D:\BBS-09-01-2026> python test_core_registry.py

{'status': 'CORE_REGISTERED', 'core_id': 'core_01'}

{'status': 'CORE_REGISTERED', 'core_id': 'core_02'}

{'status': 'CORE_REGISTERED', 'core_id': 'core_03'}

{'status': 'CORE_UPDATED'}

{'total_cores': 3, 'cores': [{'core_id': 'core_01', 'public_key': 'pubkey_01', 'status':
'ACTIVE', 'state_root': 'root_example', 'last_seen': '2026-03-05T20:05:45.621902+00:00'},
{'core_id': 'core_02', 'public_key': 'pubkey_02', 'status': 'ACTIVE', 'state_root': None,
'last_seen': '2026-03-05T20:05:45.588636+00:00'}, {'core_id': 'core_03', 'public_key':
'pubkey_03', 'status': 'ACTIVE', 'state_root': None, 'last_seen': '2026-03-
05T20:05:45.605108+00:00'}]}

PS D:\BBS-09-01-2026> python test_core_handshake.py

{'core_id': 'core_04', 'public_key': 'pubkey_04', 'state_root': 'state_root_example',
'handshake_time': '2026-03-05T20:11:55.734671+00:00', 'status': 'HANDSHAKE_ACCEPTED'}

PS D:\BBS-09-01-2026> python test_consensus_engine.py

{'status': 'CONSENSUS_REACHED', 'state_root': 'root_A', 'votes': 3, 'total_cores': 4}

PS D:\BBS-09-01-2026> python test_fork_detector.py

{'status': 'RED_FORK_DETECTED', 'fork_roots': {'root_A': 2, 'root_B': 2}}

PS D:\BBS-09-01-2026> python test_state_canonizer.py

{'status': 'STATE_CANONIZED', 'canonical_root': 'root_A', 'votes': 3, 'total_cores': 4}

PS D:\BBS-09-01-2026> python test_consensus_logger.py

{'status': 'CONSENSUS_LOGGED', 'canonical_root': 'root_A'}

PS D:\BBS-09-01-2026> python test_octopus_network.py

{'network_status': 'FORK_RESOLVED', 'canonical_root': 'root_A', 'votes': 3}

PS D:\BBS-09-01-2026> dir epistemic_v1_storage



    Directory: D:\BBS-09-01-2026\epistemic_v1_storage



Mode                 LastWriteTime         Length Name

----                 -------------         ------ ----

d-----        22.02.2026     16:32                truths

d-----        22.02.2026     16:32                index

d-----        22.02.2026     16:32                state_roots

d-----        22.02.2026     17:20                identity

-a----        04.03.2026     21:44           3844 epistemic_snapshot.json

-a----        05.03.2026     20:00            631 recovery_events.jsonl
```

```
PS D:\BBS-09-01-2026> dir epistemic_network_v1


    Directory: D:\BBS-09-01-2026\epistemic_network_v1


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        02.03.2026      8:00                node_A
d-----        02.03.2026      8:00                node_B
d-----        02.03.2026      8:00                node_C
d-----        02.03.2026      8:00                network_state
d-----        02.03.2026      8:00                node_status
-a----        05.03.2026     21:05            532 core_registry.json
-a----        05.03.2026     21:37            137 consensus_events.jsonl


PS D:\BBS-09-01-2026> python test_core_identity.py
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_01'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_02'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_03'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_04'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_05'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_06'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_07'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_08'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_09'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_10'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_11'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_12'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_13'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_14'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_15'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_16'}
{'status': 'CORE_IDENTITY_CREATED', 'core_id': 'core_17'}
PS D:\BBS-09-01-2026> cd D:\BBS-09-01-2026
PS D:\BBS-09-01-2026> python system_full_audit.py
SYSTEM AUDIT COMPLETE
```

```
TREE: D:\BBS-09-01-2026\SYSTEM_TREE_AUDIT.txt

FILE INDEX: D:\BBS-09-01-2026\SYSTEM_FILE_INDEX_AUDIT.csv

REPORT: D:\BBS-09-01-2026\SYSTEM_INVENTORY_AUDIT.json

PS D:\BBS-09-01-2026> python test_epistemic_classifier.py

{'event_type': 'anomaly', 'domain': 'finance', 'timestamp': '2026-03-06T06:31:12.720763',
'payload': {'type': 'anomaly', 'domain': 'finance', 'contract_id': 'PROC-77821',
'expected_price': 3200000, 'observed_price': 8700000, 'severity': 'HIGH', 'confidence': 0.92},
'reference_event_id': None, 'severity': 'HIGH', 'confidence': 0.92}

PS D:\BBS-09-01-2026> ^C

PS D:\BBS-09-01-2026> python test_epistemic_ingestion.py

{'status': 'EVENT_RECORDED', 'event_hash':
'8044a7cb22114e7e2d18648aa40a8b9be26c0b9e251386d5c7d22c8f6ef797ba'}

PS D:\BBS-09-01-2026> python test_state_commit.py

{'status': 'STATE_COMMITTED', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae', 'event_count': 1}

PS D:\BBS-09-01-2026> python test_state_broadcast.py

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_01', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

PS D:\BBS-09-01-2026> test_octopus_network_full.py

test_octopus_network_full.py : The term 'test_octopus_network_full.py' is not recognized as the
name of a cmdlet, funct

ion, script file, or operable program. Check the spelling of the name, or if a path was included,
verify that the path

is correct and try again.

At line:1 char:1

+ test_octopus_network_full.py

+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

    + CategoryInfo          : ObjectNotFound: (test_octopus_network_full.py:String) [],
CommandNotFoundException

    + FullyQualifiedErrorId : CommandNotFoundException



Suggestion [3,General]: The command test_octopus_network_full.py was not found, but does exist in
the current location. Windows PowerShell does not load commands from the current location by
default. If you trust this command, instead type: ".\test_octopus_network_full.py". See "get-help
about_Command_Precedence" for more details.

PS D:\BBS-09-01-2026> python test_octopus_network_full.py

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_01', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_02', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_03', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_04', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}
```

```
{'status': 'ROOT_BROADCASTED', 'core_id': 'core_05', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_06', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_07', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_08', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_09', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_10', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_11', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_12', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_13', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_14', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_15', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_16', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}

{'status': 'ROOT_BROADCASTED', 'core_id': 'core_17', 'merkle_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae'}


CONSENSUS RESULT

{'status': 'CONSENSUS_RESULT', 'canonical_root':
'd06ead5e6d8fba3b632a997d642da87eb662fb50de8e602783a6c0984003c0ae', 'votes': 18, 'total_cores':
18}

PS D:\BBS-09-01-2026> python test_anomaly_engine.py

{'signals_detected': 1, 'signals': [{'type': 'PRICE_ANOMALY', 'severity': 'HIGH', 'deviation':
2.71875}]}

PS D:\BBS-09-01-2026> python test_contradiction_engine.py

{'contradictions_detected': 1, 'contradictions': [{'type': 'PRICE_INCONSISTENCY', 'severity':
'HIGH'}]}

PS D:\BBS-09-01-2026> test_signal_engine.py

test_signal_engine.py : The term 'test_signal_engine.py' is not recognized as the name of a
cmdlet, function, script fi

le, or operable program. Check the spelling of the name, or if a path was included, verify that
the path is correct and

 try again.

At line:1 char:1

+ test_signal_engine.py

+ ~~~~~~~~~~~~~~~~~~~~
```

```
    + CategoryInfo          : ObjectNotFound: (test_signal_engine.py:String) [],
CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException



Suggestion [3,General]: The command test_signal_engine.py was not found, but does exist in the
current location. Windows PowerShell does not load commands from the current location by default.
If you trust this command, instead type: ".\test_signal_engine.py". See "get-help
about_Command_Precedence" for more details.
PS D:\BBS-09-01-2026> python test_signal_engine.py
{'alerts_generated': 2, 'alerts': [{'signal_type': 'ANOMALY_ALERT', 'details': {'type':
'PRICE_ANOMALY', 'severity': 'HIGH'}}, {'signal_type': 'CONTRADICTION_ALERT', 'details': {'type':
'PRICE_INCONSISTENCY', 'severity': 'HIGH'}}]}
PS D:\BBS-09-01-2026> python test_truth_pipeline.py
{'event': {'event_type': 'anomaly', 'domain': 'finance', 'timestamp': '2026-03-
07T15:04:52.903186', 'payload': {'type': 'anomaly', 'domain': 'finance', 'expected_price':
3200000, 'observed_price': 8700000, 'confidence': 0.92}, 'reference_event_id': None, 'severity':
None, 'confidence': 0.92}, 'anomaly': {'signals_detected': 1, 'signals': [{'type':
'PRICE_ANOMALY', 'severity': 'HIGH', 'deviation': 2.71875}]}, 'contradictions':
{'contradictions_detected': 0, 'contradictions': []}, 'signals': {'alerts_generated': 1,
'alerts': [{'signal_type': 'ANOMALY_ALERT', 'details': {'type': 'PRICE_ANOMALY', 'severity':
'HIGH', 'deviation': 2.71875}}]}, 'ingestion': {'status': 'EVENT_RECORDED', 'event_hash':
'636603fe8d5e78b75564738e0059b30fa421217336ec0133cc3a5432b716ab0b'}, 'state': {'status':
'STATE_COMMITTED', 'merkle_root':
'c8de0177c51a9eae48c323f23f831b149b8f3092023881a4c9ae118497a84c15', 'event_count': 2}}
PS D:\BBS-09-01-2026> python test_response_engine.py
{'responses_generated': 1, 'responses': [{'response': 'CRITICAL', 'action':
'RAISE_SYSTEM_ALERT'}]}
PS D:\BBS-09-01-2026> python test_cross_domain_engine.py
{'cross_domain_signals': 2, 'signals': [{'type': 'CROSS_DOMAIN_INTERACTION', 'domains':
['medicine', 'law', 'finance']}, {'type': 'SYSTEMIC_CORRUPTION_RISK', 'severity': 'HIGH'}]}
PS D:\BBS-09-01-2026> python test_domain_core_registry.py
{'total_cores': 5, 'cores': [{'core_id': 'core_01', 'domain': 'epistemic', 'public_key': None,
'status': 'ACTIVE', 'created_at': '2026-03-07T15:58:46.916503'}, {'core_id': 'core_02', 'domain':
'finance', 'public_key': None, 'status': 'ACTIVE', 'created_at': '2026-03-07T15:58:46.916534'},
{'core_id': 'core_03', 'domain': 'law', 'public_key': None, 'status': 'ACTIVE', 'created_at':
'2026-03-07T15:58:46.916539'}, {'core_id': 'core_04', 'domain': 'medicine', 'public_key': None,
'status': 'ACTIVE', 'created_at': '2026-03-07T15:58:46.916543'}, {'core_id': 'core_05', 'domain':
'education', 'public_key': None, 'status': 'ACTIVE', 'created_at': '2026-03-
07T15:58:46.916547'}]}
PS D:\BBS-09-01-2026> python test_core_sync_engine.py
{'cores': 3, 'unique_roots': 2, 'state_consistent': False}
{'status': 'STATE_SYNCHRONIZED', 'source_core': 'core_01', 'state_root': 'root_A',
'synced_cores': 2}
{'cores': 3, 'unique_roots': 1, 'state_consistent': True}
PS D:\BBS-09-01-2026> python test_domain_interaction_engine.py
{'interaction_detected': True, 'source_domain': 'finance', 'affected_domains': ['law',
'medicine', 'education']}
PS D:\BBS-09-01-2026> python test_systemic_risk_engine.py
{'systemic_risk_detected': True, 'anomaly_count': 3, 'domains_involved': ['medicine', 'finance',
'law'], 'severity': 'HIGH'}
```

```
PS D:\BBS-09-01-2026> python test_epistemic_audit_logger.py

{'status': 'LOG_RECORDED', 'file': 'epistemic_v1_storage\\epistemic_analysis_log.jsonl'}

PS D:\BBS-09-01-2026> python test_institutional_alert_engine.py

{'status': 'INSTITUTIONAL_ALERT_CREATED', 'alert_level': 'CRITICAL', 'file':
'epistemic_v1_storage\\institutional_alerts.jsonl'}

PS D:\BBS-09-01-2026> python test_institution_routing_engine.py

{'institutions_notified': ['STATE_AUDIT_SERVICE', 'CONSTITUTIONAL_COURT', 'MINISTRY_OF_FINANCE',
'ANTI_CORRUPTION_BUREAU', 'MINISTRY_OF_JUSTICE', 'MINISTRY_OF_HEALTH', 'NATIONAL_HEALTH_FUND'],
'count': 7}

PS D:\BBS-09-01-2026> python test_institution_response_tracker.py

{'status': 'RESPONSE_RECORDED', 'institution': 'MINISTRY_OF_FINANCE'}

PS D:\BBS-09-01-2026> python test_resolution_engine.py

{'alert_id': 'ALERT_001', 'resolved': False, 'responses': 1}

PS D:\BBS-09-01-2026> python test_case_engine.py

{'status': 'CASE_CREATED', 'case_id': 'CASE_4ac321f3a93a'}

{'status': 'CASE_UPDATED', 'case_id': 'CASE_4ac321f3a93a'}

PS D:\BBS-09-01-2026> python test_archive_manager.py

{'status': 'ARCHIVE_COMPLETE', 'moved_records': 1}

PS D:\BBS-09-01-2026> python test_leo_system_chronicle.py

{'status': 'CHRONICLE_RECORDED', 'component': 'CASE_ENGINE'}

PS D:\BBS-09-01-2026> python test_leo_full_system_pipeline.py

Traceback (most recent call last):

  File "D:\BBS-09-01-2026\test_leo_full_system_pipeline.py", line 9, in <module>

    from leo.runtime.epistemic.classifier.epistemic_classifier import EpistemicClassifier

ModuleNotFoundError: No module named 'leo.runtime.epistemic.classifier'

PS D:\BBS-09-01-2026> python test_leo_full_system_pipeline.py

Traceback (most recent call last):

  File "D:\BBS-09-01-2026\test_leo_full_system_pipeline.py", line 9, in <module>

    from leo.runtime.epistemic.classifier import EpistemicClassifier

ModuleNotFoundError: No module named 'leo.runtime.epistemic.classifier'

PS D:\BBS-09-01-2026> dir leo\runtime\epistemic



    Directory: D:\BBS-09-01-2026\leo\runtime\epistemic



Mode                 LastWriteTime         Length Name

----                 -------------         ------ ----

d-----        16.02.2026      7:10                __pycache__

d-----        21.02.2026     19:23                v0_legacy
```

```
d-----        21.02.2026     19:32                ingestion_v1
d-----        22.02.2026     18:53                network_v1
d-----        05.03.2026     19:47                recovery_v1
d-----        05.03.2026     20:18                network_rebuild_v1
d-----        05.03.2026     21:04                network_v2
-a----        20.02.2026     23:09           3683 ingestion.py
-a----        07.03.2026     17:52            807 epistemic_audit_logger.py


PS D:\BBS-09-01-2026> python test_leo_full_system_pipeline.py
Traceback (most recent call last):
  File "D:\BBS-09-01-2026\test_leo_full_system_pipeline.py", line 9, in <module>
    from leo.runtime.analysis.epistemic_classifier import EpistemicClassifier
ModuleNotFoundError: No module named 'leo.runtime.analysis.epistemic_classifier'
PS D:\BBS-09-01-2026> python test_leo_full_system_pipeline.py
Traceback (most recent call last):
  File "D:\BBS-09-01-2026\test_leo_full_system_pipeline.py", line 14, in <module>
    from leo.runtime.epistemic.ingestion import IngestionEngine
ImportError: cannot import name 'IngestionEngine' from 'leo.runtime.epistemic.ingestion' (D:\BBS-09-01-2026\leo\runtime\epistemic\ingestion.py)
PS D:\BBS-09-01-2026> type leo\runtime\epistemic\ingestion.py
import hashlib
import uuid
from datetime import datetime
from typing import Dict, Any


class ValidationError(Exception):
    pass


class IntegrityError(Exception):
    pass


class AdmissionResult:
    def __init__(self, status: str, reason: str = None):
        self.status = status
        self.reason = reason
```

```
class TruthIngestionEngine:
    """
    Runtime-bound implementation of TRUTH_INGESTION_PIPELINE_v1.0
    """

    def ingest(self, raw_input: Dict[str, Any]) -> AdmissionResult:
        validated = self.validate(raw_input)
        content_hash = self.compute_hash(validated["content"])
        state_anchor = self.compute_state_anchor()

        cto = self.build_cto(validated, content_hash, state_anchor)

        decision = self.make_decision(cto)

        if decision == "ADMITTED":
            self.commit(cto, validated["content"])
            return AdmissionResult(status="ADMITTED")

        elif decision == "ESCALATED":
            self.log_escalation(cto)
            return AdmissionResult(status="ESCALATED")

        else:
            return AdmissionResult(status="REJECTED", reason="Validation failure")

    # ----------------------------
    # Validation
    # ----------------------------

    def validate(self, raw_input: Dict[str, Any]) -> Dict[str, Any]:
        required_fields = ["content", "source_type", "source_identity", "semantic_type"]

        for field in required_fields:
            if field not in raw_input:
                raise ValidationError(f"Missing required field: {field}")

        if not raw_input["content"]:
```

```
            raise ValidationError("Content cannot be empty")

        return raw_input

    # ----------------------------
    # Hashing
    # ----------------------------

    def compute_hash(self, content: str) -> str:
        normalized = content.strip().encode("utf-8")
        return hashlib.sha256(normalized).hexdigest()

    def compute_state_anchor(self) -> str:
        # Placeholder for real state hash logic
        now = datetime.utcnow().isoformat()
        return hashlib.sha256(now.encode()).hexdigest()

    # ----------------------------
    # CTO Builder
    # ----------------------------

    def build_cto(self, validated: Dict[str, Any], content_hash: str, state_anchor: str) ->
Dict[str, Any]:
        return {
            "truth_id": str(uuid.uuid4()),
            "content_hash": content_hash,
            "source_type": validated["source_type"],
            "source_identity": validated["source_identity"],
            "timestamp_utc": datetime.utcnow().isoformat(),
            "semantic_type": validated["semantic_type"],
            "version": "1.0.0",
            "parent_truth": validated.get("parent_truth"),
            "state_anchor_hash": state_anchor,
            "admission_signature": "SYSTEM_SIGNATURE_PLACEHOLDER"
        }

    # ----------------------------
    # Decision Logic
    # ----------------------------
```

```
    def make_decision(self, cto: Dict[str, Any]) -> str:
        # Placeholder deterministic decision logic
        if not cto["content_hash"]:
            return "REJECTED"

        return "ADMITTED"

    # ----------------------------
    # Commit / Escalation
    # ----------------------------

    def commit(self, cto: Dict[str, Any], content: str):
        # TODO: integrate with EpistemicMemory atomic_commit()
        print("Committing truth:", cto["truth_id"])

    def log_escalation(self, cto: Dict[str, Any]):
        # TODO: integrate with Governance module
        print("Escalation required for truth:", cto["truth_id"])
PS D:\BBS-09-01-2026> python test_leo_full_system_pipeline.py
Traceback (most recent call last):
  File "D:\BBS-09-01-2026\test_leo_full_system_pipeline.py", line 16, in <module>
    from leo.runtime.state.state_commit import StateCommit
ModuleNotFoundError: No module named 'leo.runtime.state.state_commit'
PS D:\BBS-09-01-2026> python test_leo_full_system_pipeline.py
Traceback (most recent call last):
  File "D:\BBS-09-01-2026\test_leo_full_system_pipeline.py", line 24, in <module>
    from leo.runtime.network.consensus_engine import ConsensusEngine
ModuleNotFoundError: No module named 'leo.runtime.network.consensus_engine'
PS D:\BBS-09-01-2026>  python test_leo_full_system_pipeline.py
Traceback (most recent call last):
  File "D:\BBS-09-01-2026\test_leo_full_system_pipeline.py", line 24, in <module>
    from leo.runtime.epistemic.network_v2.consensus_engine import ConsensusEngine
ImportError: cannot import name 'ConsensusEngine' from
'leo.runtime.epistemic.network_v2.consensus_engine' (D:\BBS-09-01-
2026\leo\runtime\epistemic\network_v2\consensus_engine.py)
PS D:\BBS-09-01-2026> python test_leo_full_system_pipeline.py


--- LEO FULL PIPELINE START ---
```

```
EVENT CREATED

EVENT CLASSIFIED

ANOMALY DETECTION COMPLETE

CONTRADICTION CHECK COMPLETE

SIGNALS GENERATED

Committing truth: 904b0ea2-d180-4929-88a4-ea8091d4745e

EVENT STORED

STATE COMMITTED

INSTITUTIONS ROUTED

INSTITUTION RESPONSE RECORDED

RESOLUTION CHECK COMPLETE

CASE CREATED

CONSENSUS RESULT

{'status': 'CONSENSUS_REACHED', 'state_root':
'fa4e287ab537c5f0858f7b63d4f7075e03a42150cf1113bbf8c023cc3533b147', 'votes': 3, 'total_cores': 3}


--- LEO PIPELINE COMPLETE ---


PS D:\BBS-09-01-2026>
```

3. System Module Inventory

As shown in the audit, the LEO runtime is composed of specialized engines for:

- **Epistemic Logic:** truth_ingestion_engine_v1.py
- **Anomaly Detection:** epistemic_anomaly_engine.py
- **Consensus:** consensus_engine_v1.py
- **Graph Mapping:** epistemic_graph_engine.py